

SPARSH BANSAL

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Tempe, Arizona

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SUMMARY

Software Engineer (MS CS @ ASU, GPA 4.0) with hands-on experience designing and deploying cloud-native microservices, building CI/CD pipelines, and developing ML-powered applications. Skilled in Java/Spring Boot and Python, with proven success reducing manual engineering effort by 60% at Cummins and deploying scalable backend services at Credit One Bank. Passionate about building secure, data-driven, and high-performance systems at scale.

SKILLS

Languages & Databases: Java, Python, JavaScript/TypeScript, SQL (PostgreSQL, MySQL), HTML/CSS, C,C++
Backend & Frameworks: Spring Boot, Node.js, GraphQL, Prisma, Microservices, REST APIs, Kafka, JUnit
DevOps & Cloud: AWS (EKS, RDS, S3, Lambda), Docker, Kubernetes, Jenkins (CI/CD), Git, Azure, GCP
Frontend & API Tools: React, Next.js, Tailwind CSS, React, Apigee, Okta, Postman, OAuth 2.0

EDUCATION

Arizona State University

Master of Science in Computer Science

GPA: 4.0

January 2023 – December 2025

Tempe, AZ

Virginia Tech University

Bachelor of Science in Computer Science

August 2019 – December 2022

Blacksburg, VA

PROFESSIONAL EXPERIENCE

Software Quality Engineer

Cummins Inc.

January 2024 –August 2024

Columbus, IN

- Automated engineering record correlation by developing an AI-driven workflow, which replaced manual documentation, **cutting effort by 60%**.
- Built and deployed interactive **Power BI dashboards** to track issue resolution metrics, which improved stakeholder visibility and accelerated decision-making.
- Coordinated **process standardization across 5 divisions**, introducing regulatory updates that **reduced turnaround time by 27%** while maintaining **100% compliance**.

Full-Stack Developer Intern

Credit One Bank

June 2023 – August 2023

Las Vegas, NV

- Developed a **Spring Boot microservice** to handle CRUD operations for internal student profile data, which improved backend scalability and supported business planning tools.
- Built a **CI/CD pipeline in Jenkins** to automate software delivery across multi-cloud and on-prem environments, which streamlined deployments and reduced release errors.
- Secured APIs by implementing **Apigee proxies with key-based protection** and integrated **Okta SSO**, which strengthened data security and compliance standards.
- Enhanced website authentication by applying **OAuth 2.0 standards**, which improved login reliability and reduced friction in the user experience.

ACADEMIC EXPERIENCE

AI-Powered Collaboration Platform

January 2025 – May 2025

- Built a full-stack collaboration platform with real-time document editing and team chat using **React, Next.js, and TypeScript**, enabling seamless multi-user productivity.
- Engineered a scalable backend with **Node.js, GraphQL, and Prisma on PostgreSQL (pgvector)** to power semantic search and knowledge retrieval.
- Integrated **LLM-based features (LangChain + open-source models)** for auto-summarization and intelligent Q&A, reducing manual knowledge curation for users.
- Deployed cloud-native infrastructure on **AWS (EKS, RDS, S3)** with **Docker, Kubernetes, and Terraform**, ensuring high availability and automated CI/CD.

Truthkeep: Advanced Disinformation Detection

August 2023 – May 2024

- Led a **team of four** to design a disinformation detection platform, which processed **500K+ articles**, enabling large-scale trend analysis.
- Built scalable **data pipelines using Apache Airflow + BeautifulSoup** for automated scraping and ingestion, which increased processing efficiency by **41%**.
- Applied **Logistic Regression & Random Forest classifiers** with TF-IDF and LDA feature extraction, which achieved **88% accuracy** in identifying unreliable sources.